

$$\frac{dV}{dt} = -\frac{B^2 l^2 v m}{R}$$

Electromagnetism II

$$\frac{dV}{dt} = -\frac{B^2 l^2 v m}{R}$$

$$\frac{dV}{dt} = \ln V(t) = -\frac{B^2 l^2 v m}{R} t$$

$$V(t) = e^{-\frac{B^2 l^2 v m}{R} t}$$

2023(8030)

Andrew

Trevanna

Midterm Exam

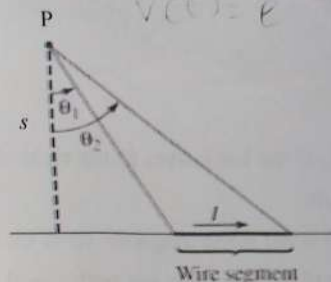
Intermediate examination

(The meanings of the variables and constants are the same as those explained in the lecture.)

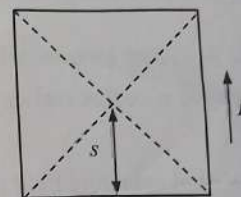
1. A constant current I is flowing in a straight wire. The magnitude of magnetic field B at point P is given as follows using θ_1 and θ_2 in the figure.

$$B = \frac{\mu_0 I}{4\pi s} (\sin \theta_2 - \sin \theta_1)$$

s is the distance of P from the wire. Use this result for the following questions.



- (1) Find the magnetic field B at the center of a square loop. Let s be the distance from the center to the sides.
- (2) Find B at the center of a regular n -sided polygon. Let s be the distance from the center to any side.
- (3) Find B at the center of a circular loop in the limit $n \rightarrow \infty$.



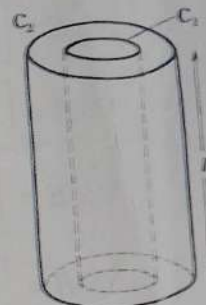
2. The magnetic field B generated by a solenoid with n turns per unit length carrying a steady current I is given as follows.

$$B = \mu_0 n I \quad (\text{inside the solenoid})$$

$$0 \quad (\text{outside the solenoid})$$

Use the result for the following questions.

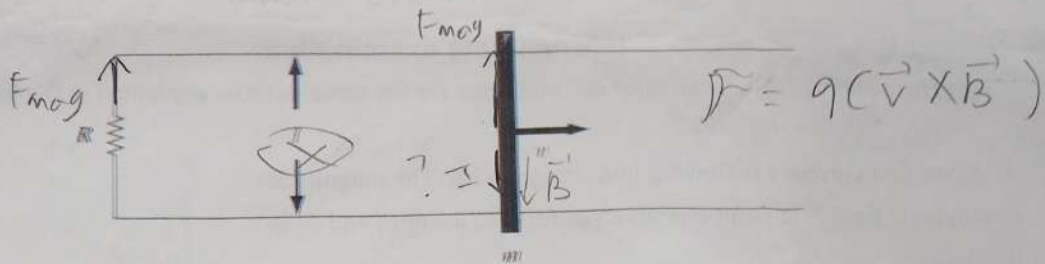
- (1) There are two long coaxial solenoids C_1 and C_2 of a length l , with n_1 and n_2 turns per unit length, respectively. Current I_1 and I_2 are flowing in the opposite directions within C_1 and C_2 , respectively. Find the magnetic field B_1 within solenoid C_1 .
- (2) Find the magnetic field B_2 in the region between the two solenoids.
- (3) Find the magnetic field B_3 outside of solenoid C_2 .



3. Answer the following questions.

- (1) Explain "paramagnetism" briefly.
- (2) Explain "diamagnetism" briefly.
- (3) Explain "ferromagnetism" briefly.

4. A metal bar of mass m slides frictionlessly on two parallel conducting rails a distance l apart. A resistor R is connected, and a uniform magnetic field B , pointing into the page, is applied in the entire region. Answer the following questions.



(1) If the bar moves to the right at velocity v , what is the generated current I ? In what direction does it flow?

(2) What is the magnetic force on the bar? In what direction?

(3) If the bar starts out with speed v_0 at time $t = 0$, what is its speed at a later time t ?

(4) Find the energy delivered to the resistor.

$$U = \frac{1}{2} m v^2_0$$

5. A square loop of wire (side a) lies at a distance s from a long straight wire that carries a current I . Answer the following questions.

(1) Find magnetic field the current generates as a function of s .

(2) Find the magnetic flux Φ that passes through the loop.

(3) If the loop is pulled directly away from the wire at velocity v , how about $\mathcal{E}$? In what direction (clockwise or counterclockwise) does the current flow?

(4) Find $\mathcal{E}$ when the loop is pulled to the right at velocity v ?

6. Write the right sides of Maxwell's equations in a vacuum below using variables and constants including E , B , ϵ_0 , μ_0 , ρ , and J .

$$\nabla \cdot E = \text{(a)}$$

$$\nabla \cdot B = \text{(b)}$$

$$\nabla \times E = \text{(c)}$$

$$\nabla \times B = \text{(d)}$$

$$\frac{dv}{dt} = -\frac{B^2 \ell^2 v m}{R}$$

$$\int \frac{dv}{v} = -\frac{B^2 \ell^2 m}{R} \int dt$$

$$\ln v = -\frac{B^2 \ell^2 m t}{R}$$

$$v(t) = e^{-\frac{B^2 \ell^2 m t}{R}}$$

$$v(t) = v_0 e^{-\frac{B^2 \ell^2 m t}{R}}$$

